OBJECT-ORIENTED PROGRAMMING WITH JAVA

Project Assignment

Project Title

Design and Implementation of a Software Application Using Object-Oriented Programming Principles in Java

Project Description

Students are required to design, develop, test, and document a software application of their choice using Java programming language and Object-Oriented Programming (OOP) principles.

Students may select a suitable problem from any field, provided that the proposed application demonstrates the required OOP concepts and meets the project requirements.

Project Objectives

By the end of the project, students should be able to:

1. Apply fundamental Object-Oriented Programming principles in Java.
2. Design and implement classes and objects.
3. Apply encapsulation, inheritance, polymorphism, and abstraction.
4. Implement constructors, methods, interfaces, and abstract classes.
5. Use arrays, collections, and appropriate data structures.
6. Implement exception handling and input validation.
7. Apply file handling and database connectivity where applicable.
8. Design, test, debug, and document a Java application.
9. Demonstrate problem-solving, programming, and software development skills.

Project Requirements

The application must demonstrate the following concepts and features:

1. **Classes and Objects:** Define appropriate classes and create objects to represent entities within the application.
2. **Encapsulation:** Apply access modifiers and provide appropriate methods for accessing and modifying object data.
3. **Constructors and Methods:** Implement constructors and methods to initialize objects and perform operations.
4. **Inheritance:** Demonstrate inheritance by creating suitable parent and child classes.
5. **Polymorphism:** Implement method overloading, method overriding, or other appropriate forms of polymorphism.
6. **Abstraction:** Demonstrate abstraction using abstract classes and/or interfaces.
7. **Collections and Data Structures:** Use arrays, ArrayList, or other appropriate data structures to manage information.
8. **Exception Handling:** Handle invalid inputs and unexpected runtime errors.
9. **Data Persistence:** Implement file handling or database connectivity, depending on the concepts covered in the course.
10. **User Interface:** Develop a menu-driven console application or graphical user interface.

11. **Testing:** Test the application to verify that its functions operate correctly.
12. **Documentation:** Provide appropriate technical documentation and instructions for using the application.

Project Stages

Stage 1: Project Proposal and Requirements Analysis

- Identify a problem to solve.
- Define the project objectives and scope.
- Identify the intended users.
- Specify functional and non-functional requirements.
- Submit a project proposal.

Stage 2: Classes, Objects, and Constructors

- Identify the major classes required.
- Define class attributes and methods.
- Implement constructors.
- Create objects and demonstrate their behaviour.

Stage 3: Encapsulation and Data Validation

- Apply access modifiers.
- Implement getters and setters where appropriate.
- Validate user input and object data.
- Prevent invalid data from being processed.

Stage 4: Inheritance and Polymorphism

- Design appropriate inheritance relationships.
- Implement parent and child classes.
- Demonstrate method overriding and method overloading.
- Explain how polymorphism is applied in the application.

Stage 5: Abstraction, Interfaces, and Collections

- Implement abstract classes and interfaces where appropriate.
- Apply abstraction to improve software design.
- Use collections to store and manipulate objects.
- Implement relevant searching, updating, and deletion operations.

Stage 6: Exception Handling and Data Persistence

- Implement exception handling.
- Provide meaningful error messages.
- Save and retrieve application data using files or a database.
- Test data persistence.

Stage 7: User Interface and System Testing

- Develop the application interface.
- Integrate the different modules.
- Test individual functions and the complete application.
- Identify and correct errors.

Stage 8: Final Implementation and Presentation

- Complete the application.
- Prepare the final project report.
- Submit the source code and supporting documentation.
- Demonstrate the application.
- Explain the OOP concepts implemented during the project defense.

Expected Deliverables

Students are expected to submit:

1. Project proposal.
2. Requirements specification.
3. System design and UML class diagram.
4. Complete Java source code.
5. A functional software application.
6. Test cases and test results.
7. Technical documentation and user manual.
8. Final project report.
9. Project demonstration and oral presentation.

Submission Guidelines

- The project may be completed individually or in groups.
- Students must demonstrate a clear understanding of the code they submit.
- All submitted work must be original, properly documented, and consistent with academic integrity requirements.
- The final application must demonstrate the practical application of the OOP concepts taught during the semester.
- A video link on the google drive should be submitted entailing your explanation of the project.